NED UNIVERSITY OF ENGINEERING & TECHNOLOGY

Centre for Multidisciplinary Postgraduate Programmes (CMPP) – NED Academy

Postgraduate Diploma in Data Science with AI

Final Examination – Spring-II-2026

Exam Date: June 13, 2026

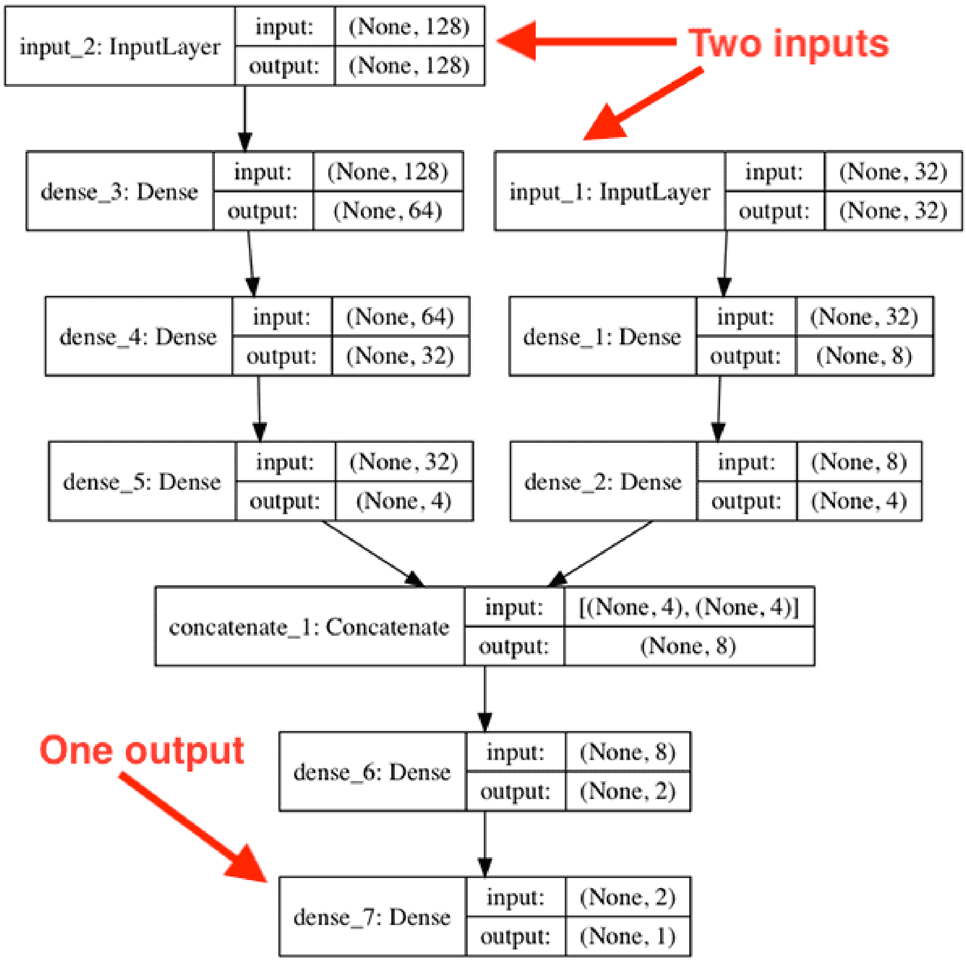
Course (Code & Title): Deep Learning

Time: 3 Hours Max. Marks: 60

Instructions:

i. Attempt all questions
ii. Marks for each question are given.
iii. You are required to abide by all rules and regulations set for the examination by the NED Academy.
iv. Total time of examination is 3 hours. No extra time will be provided after the time is over.

<table>
<tr><th>S. No.</th><th>Question</th><th>Marks</th></tr>
<tr><td>1.</td><td>Observe the given neural network diagram carefully and create the same model using the Keras Functional API. The model has two input layers and one output layer. The first input has shape (128,) and passes through three Dense layers with 64, 32, and 4 neurons. The second input has shape (32,) and passes through two Dense layers with 8 and 4 neurons. After processing both inputs separately, concatenate the outputs of both branches using a Concatenate layer. Then add two Dense layers with 2 and 1 neurons. Use a suitable activation function in the final output layer for binary classification. Finally, create the complete model using Model(inputs=[input_1, input_2], outputs=output), compile the model with a suitable optimizer and loss function, and display the model summary.


</td><td>15 Marks</td></tr>
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2.	a) Explain the difference between Image Processing and Computer Vision. Support your answer with one example of each. Marks: 3 b) An image is represented as a numerical array before it is passed to a deep learning model. Explain the role of pixels, channels, and image resolution in image representation. Marks: 3 c) Explain why image preprocessing is important before training a computer vision model. Discuss any three preprocessing techniques such as resizing, normalization, grayscale conversion, augmentation, or edge detection. Marks: 3 d) What is a pretrained model? Explain how models such as VGG16, ResNet50, or MobileNetV2 help in solving computer vision problems. Marks: 3 e) Explain the concept of Transfer Learning. Differentiate between feature extraction and fine-tuning in transfer learning. Marks: 3	Marks: 15
3.	Question: Helmet Detection Using YOLOv8 and YOLOv11 with Roboflow Dataset A construction company wants to develop an automatic safety monitoring system for its workers. The system should detect whether workers are wearing safety helmets or not. The dataset is available on Roboflow and contains annotated images with two object classes: Helmet and No Helmet. You are required to download the dataset from Roboflow and train two YOLO object detection models: YOLOv8 and YOLOv11. After training, compare both models using object detection performance metrics such as Precision, Recall, mAP@50, and mAP@50-95. Finally, perform prediction on a test image and explain how bounding boxes, class labels, and confidence scores help in helmet detection.	Marks: 15
4.	**<u>Age-Gender-Ethnicity Prediction</u>** Predict age, gender and ethnicity of people using any Pre-trained and UTKFace datasets  https://www.kaggle.com/datasets/jangedoo/utkface-new	Marks: 15